# Prime Factorization

Keep splitting a factor tree until every branch is prime.

**Name** _____ **Date** _____

### Tonight's spark

A locker tag says 84. Maya splits it into 4 × 21 and says her tree is finished. Her friend says a piece can still split. Who is right?

**Guess first: is the 4 already prime, or can it split again?**

| Number | Splits into |
|---|---|
| 84 | 4 × 21 |
| 4 | 2 × 2 |
| 21 | 3 × 7 |
| 84 | 2 × 2 × 3 × 7 |

all prime

**Our class model: the all-primes check**

### Try it

Work it out on paper. Write one answer on each line.

**1** **Which numbers at the bottom of a factor tree mean the tree is finished?**

**A.** 2, 2, 3, 5 **B.** 2, 4, 5 **C.** 3, 6, 5

**My answer** _____

**2** **Which number is composite (not prime)?**

**A.** 2 **B.** 7 **C.** 9 **D.** 11

**My answer** _____

**3** **Complete: 50 = 2 × 5 × ___**

**A.** 5 **B.** 10 **C.** 25 **D.** 2

**My answer** _____

**4** **Maya starts a tree for 48 with 6 × 8. Which numbers must she split further?**

**A.** Only 6 **B.** Only 8 **C.** Neither, because the tree is finished **D.** Both 6 and 8

**My answer** _____

**5** **Which shows 63 written with prime factors only?**

**A.** 7 × 9 **B.** 3 × 21 **C.** 3 × 3 × 7 **D.** 1 × 63

**My answer** _____

### Talk about it

Is every number at the bottom of the tree prime? How can you tell your tree is finished?

### Home mission

Write 84 on paper. Ask your student to start with 2 × 42 instead of 4 × 21. Do both trees end in the same primes?

### If we get stuck, ask

- Is every number at the bottom of the tree prime?
- Can this number be split into smaller factors?
- Do your primes multiply back to the number you started with?

**Do it on a screen**

Scan or type the link. Same lesson, with hints.

**eduwonderlab.com/lessons/6-13-part2/homework.html**

**How did it feel? (optional)**

Tricky Getting there Got it